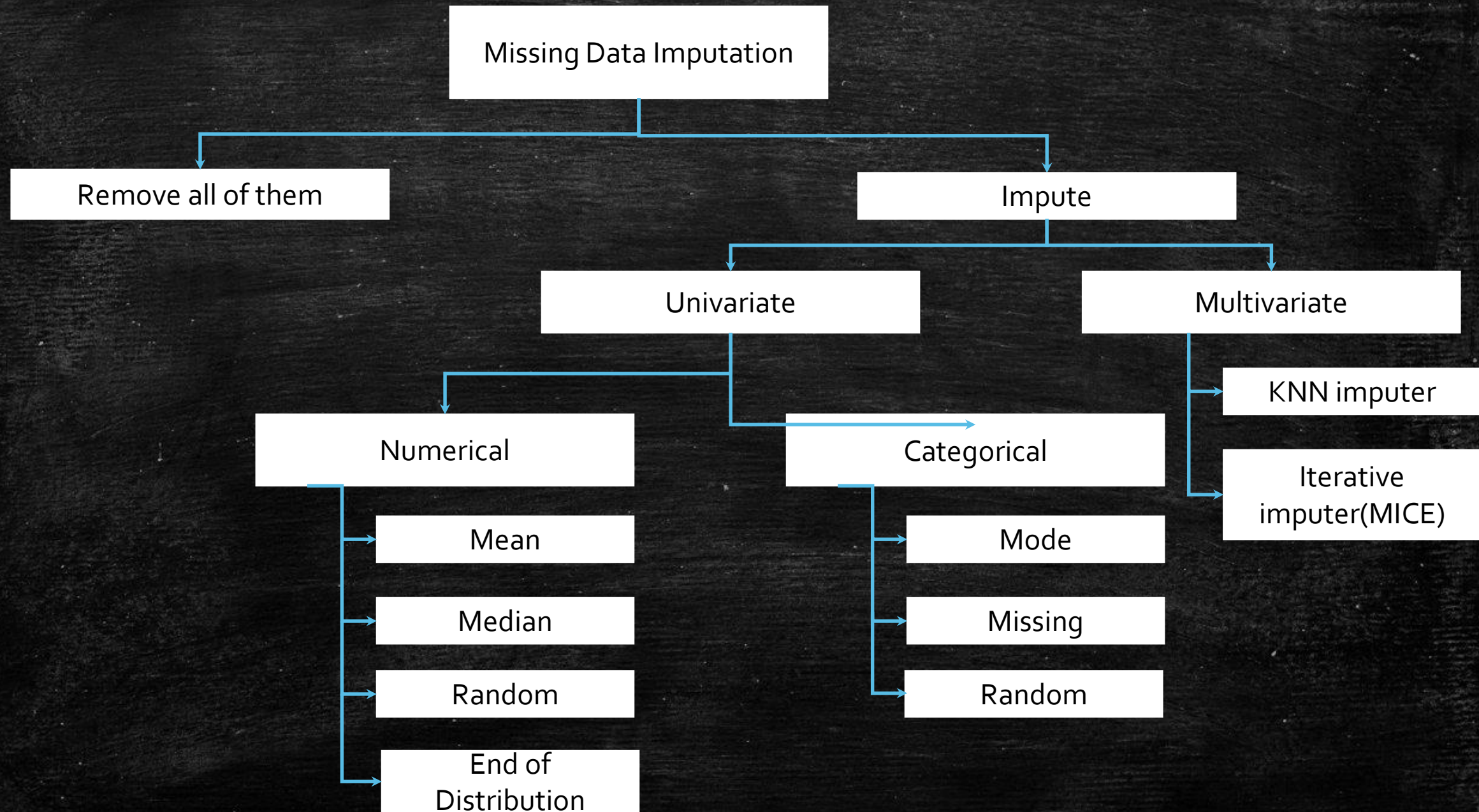




Complete Case Analysis (CCA)

- **Complete Case Analysis (CCA)**, also known as list wise deletion, is a method for handling missing values in machine learning where rows with any missing values are removed entirely from the dataset.
- CCA means analyzing only those rows from which there is information in all of the variables in the dataset.

Assumption of CCA:

Data is missing
completely at random.

Advantages of CCA

- **Simplicity:** CCA is straightforward to implement and understand. It involves removing rows with missing values, which makes the analysis process simpler and more transparent.
- **Preservation of Data Structure:** CCA retains the original structure of the dataset, as it does not require imputation or estimation of missing values.
 - This can be beneficial when preserving the integrity and interpretability of the data is important.

Disadvantages of CCA

- **Loss of Data:** It can exclude a large fraction of original dataset.
- When using our models in production, the model will not know how to handle missing data.

When to use CCA?

1. Missing Completely at Random (MCAR)
2. less than 5 % missing values , this threshold value may vary.

Missing Value
imputation for
numerical variables.

Univariate Analysis

- **Univariate analysis** is the simplest form of analyzing data.
- “Uni” means “one”, so in other words your data has only one variable.
- It doesn't deal with causes or relationships (unlike [regression](#)) and it's major purpose is to describe; It takes data, summarizes that data and finds patterns in the data.

	index	gender	age	Bodyweight	spread	offer
1	1	male	31	63.1	butter	very poor
2	2	male	73	77.5	butter	very poor
3	5	male	45	82.1	butter	very poor
4	6	male	57	61.7	butter	very poor
5	9	male	38	36.5	butter	very poor
6	11	male	27	64.0	butter	very poor
7	12	male	36	70.9	butter	very poor
8	13	male	60	70.4	butter	very poor
9	15	male	21	6	butter	very poor
10	16	male	26			
11	18	male	55			
12	22	male	27		butter	very poor
13	25	male	30	72.7	butter	very poor
14	26	male	33	77.8	butter	very poor
15	27	male	33	90.8	butter	very poor
16	28	male	58	62.4	butter	very poor
17	29	male	23	91.2	butter	very poor

Analysis of only one variable:
Univariate Analysis

Mean Median Imputation

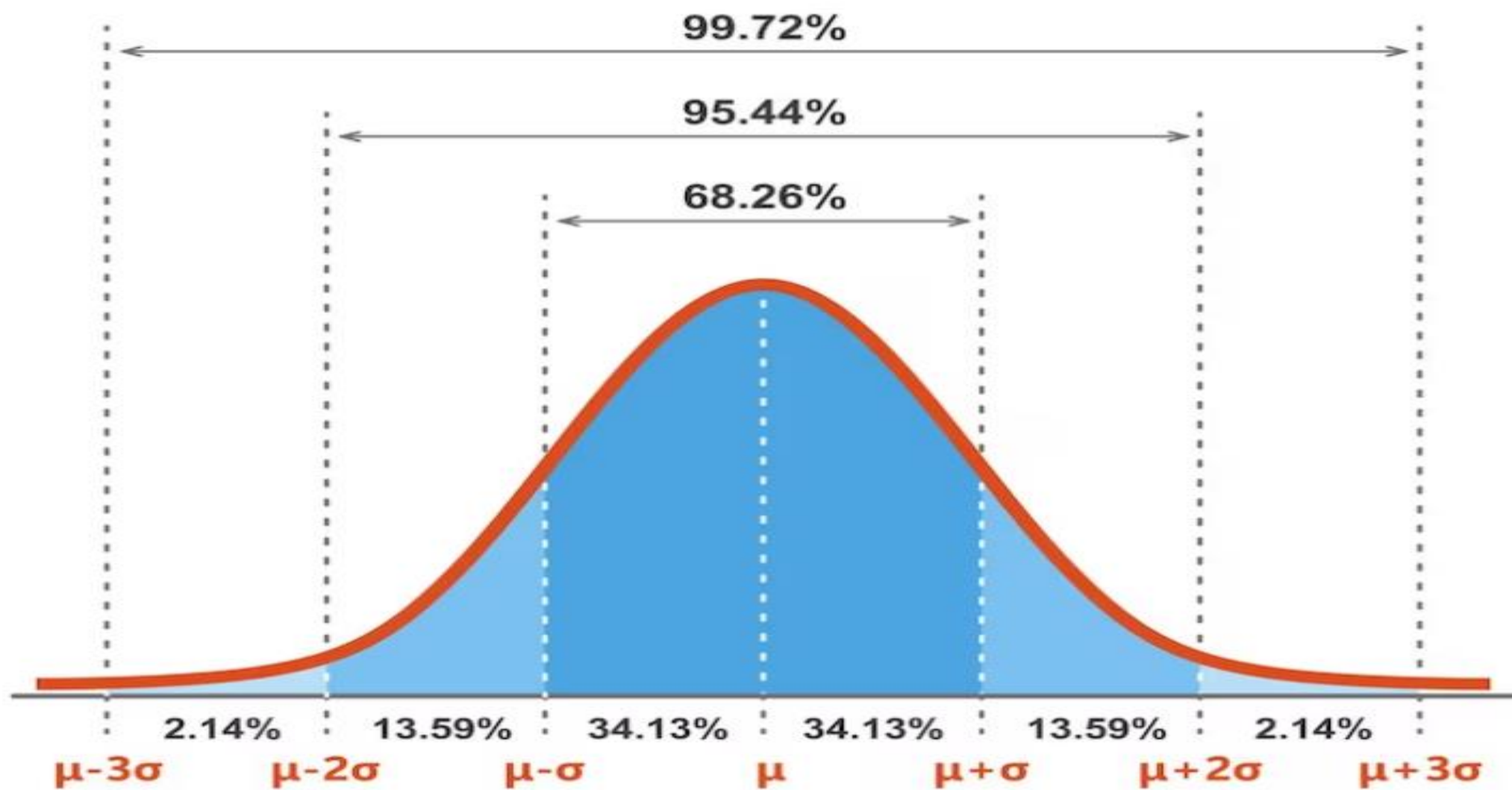
- **Mean** imputation can be used when the distribution of the data is normal, and the missing values are random.
- **Median** imputation is appropriate when the distribution of the data is skewed
- Mean and median both are used to find the **central tendency** of the data.
- Mean is affected by the **outliers** and it is not robust estimator where as median is not affected by the outlier.
- If the data is normally distributed impute data with the mean otherwise use median.

Random Value

- **Replacing Missing Value with an arbitrary number** Arbitrary number imputation consists of replacing missing values with an arbitrary value. Some commonly used values include 999, 9999, or -1 for positive distributions.
- This method is suitable for numerical variables.
- When replacing missing values with an arbitrary number, we need to be careful not to select a value close to the mean or the median, or any other common value of the distribution.
- Important: Arbitrary number imputation can be used when data is not missing at random, **when we are building non-linear models**, and when the **percentage of missing data is high**.
- **This imputation technique distorts the original variable distribution.**

Empirical Rule

- The empirical rule, also sometimes called the **three-sigma** or **68-95-99.7 rule**, is a statistical rule which states that for **normally distributed data**.
- Almost all observed data will fall within three standard deviation (σ) of the mean (μ) of the data.
- The empirical rule predicts that in normal distributions:
 - **68%** of observations fall within the first standard deviation ($\mu \pm \sigma$).
 - **95%** within the first two standard deviations ($\mu \pm 2\sigma$).
 - **99.7%** within the first three standard deviations ($\mu \pm 3\sigma$) of the mean.



The 5 number summary

- The five number summary is another name for the visual representation of the box or whisker plot.
- Five number summary consists of:
 1. The median (2nd quartile) . (Q₂)
 2. The 1st quartile. (Q₁)
 3. The 3rd quartile. (Q₃)
 4. The maximum value in a data set. (MAX)
 5. The minimum value in a data set. (MIN)

Steps for the construction of Box plot

Step 1 – Find the median.

- Remember, the median is the middle value in a data set.

18, 27, 34, 52, 54, 59, 61, 68, 78, 82, 85, 87, 91, 93, 100.

- 68 is the median of this data set.

Step 2 – Find the lower quartile.(Q1)

- The lower quartile is the median of the data set to the left of 68.

(18, 27, 34, 52, 54, 59, 61,) 68, 78, 82, 85, 87, 91, 93, 100

- Here, 52 is the lower quartile.

Or

$$Q_1 = (25/100) * (n+1)$$

$$Q_1 = (.25) * 16$$

$$Q_1 = 4$$

Step 3 – Find the upper quartile.(Q3)

- The upper quartile is the median of the data set to the right of 68.

18, 27, 34, 52, 54, 59, 61, 68, (78, 82, 85, 87, 91, 93, 100)

- 87 is the upper quartile.

Or

$$Q_3 = (75/100) * (n+1)$$

$$Q_3 = (.75) * 16$$

$$Q_3 = 12$$

Step 4 – Find the maximum and minimum values in the set.

- The maximum is the greatest value in the data set.
- The minimum is the least value in the data set.

18, 27, 34, 52, 54, 59, 61, 68, 78, 82, 85, 87, 91, 93, 100

- 18 is the minimum and 100 is the maximum.

Step 5 – Find the inter-quartile range (IQR).

- The inter-quartile (IQR) range is the difference between the upper and lower quartiles.

$$\text{IQR} = Q_3 - Q_1$$

$$\text{Upper Quartile (} Q_3 \text{)} = 87$$

$$\text{Lower Quartile (} Q_1 \text{)} = 52$$

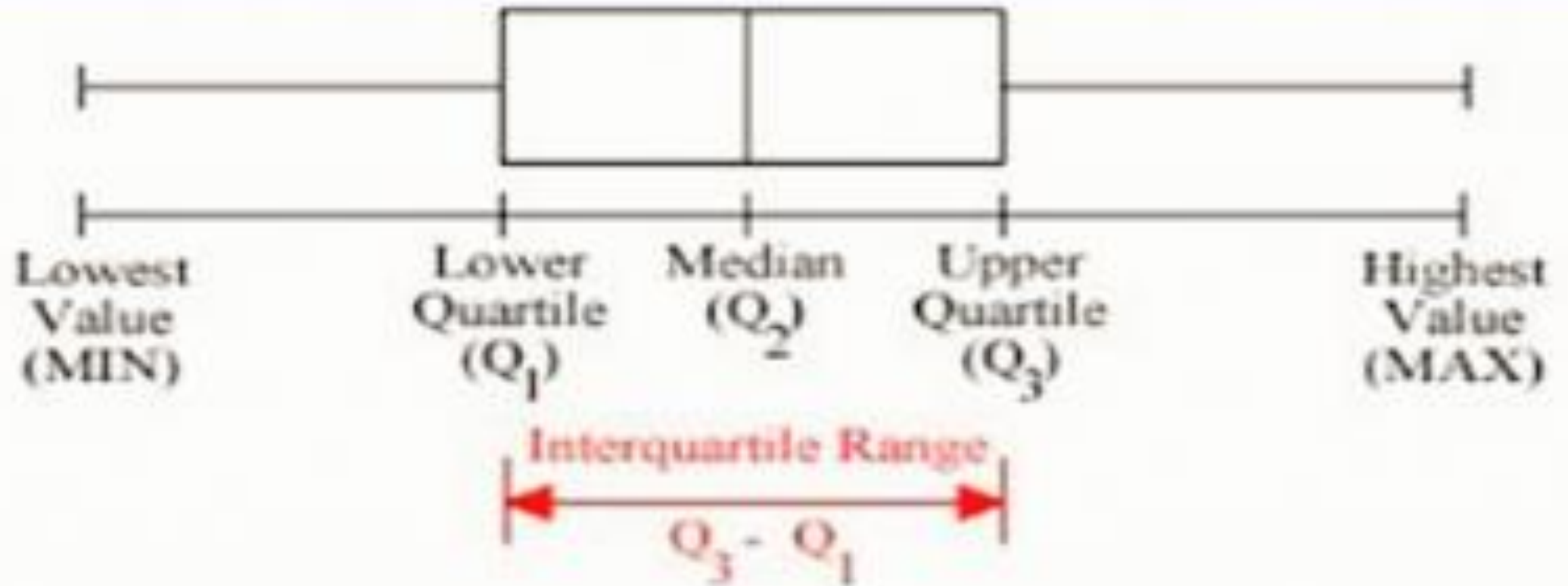
$$87 - 52 = 35$$

$$35 = \text{IQR}$$

Summary:

- The 5 number summary of above example is;
Organize the 5 number summary
- **Median = 68**
- **Lower Quartile = 52**
- **Upper Quartile = 87**
- **Max = 100**
- **Min = 18**
- **IQR = 35**

5 number summary



Normal Distribution



Positive Skew



Negative Skew



Outlier Detection using lower and higher bracket.

- Determine the values of lower and higher bracket.
- Keep only those values which are in between the lower and higher bracket.
- Remaining values which are outside of this particular bracket will be treated as an OUTLIER and it will be removed.

- Lower bracket:

$$Q_1 - 1.5 * IQR$$

- Higher bracket:

$$Q_3 + 1.5 * IQR$$

The median (Q2) tells us an average

The range tells us the spread of the data

The interquartile range tells us the spread of the middle 50% of the data

End of distribution Imputation

- According to the IQR proximity, Missing values will be replaced with a higher bracket($Q_3 + 1.5 * IQR$) at the right tail.

▪ OR

Missing values will be replaced with a lower bracket($Q_1 - 1.5 * IQR$) at the left tail.

- When to use IQR Proximity or *mean plus or minus three times the standard deviation* :
 - If the variable distribution is **skewed** we will use **IQR proximity**.
 - Else we will use **mean plus or minus three times the standard deviation** to fill the missing values. (variable follows the **normal** distribution.)

When to use End of distribution imputation:

- When data is not missing at random.
- This approach is particularly useful when the variable follows a skewed distribution or contains outliers, as it helps preserve the original shape of the distribution.

Mode Imputation

- Mode imputation is used when dealing with categorical data, where the mode represents the most common category.
- Same assumptions as of Mean/Median imputation.

Random Sample Imputation

- Random sample imputation is a technique used to impute missing values in a dataset by randomly sampling values from the non-missing entries of the same variable. Here's how it works:
 1. Identify the variable with missing values.
 2. For each missing value, randomly select a value from the non-missing values of the same variable.
 3. Assign the randomly selected value to the missing value.
- This approach preserves the overall distribution of the variable and introduces variability into the imputed values, which can be useful when other imputation methods may not be appropriate.

KNN Example

Feature 1	Feature 2	Feature 3	Feature 4
-----	33	34	23
30	43	23	-----
55	67	31	90
12	-----	56	44

$\text{dist}(x,y) = \text{sqrt}(\text{weight} * \text{sq. distance from present coordinates})$

where,

$\text{weight} = \text{Total \# of coordinates} / \text{\# of present coordinates}$



THANK YOU.